

EHG Posture Calibration — Mechanism Ranges

Sourced parameter ranges for the five posture-related mechanisms in the seated-EHG simulator, and their sourcing status as of this build.

Mechanism	Seated value / range	Source	Status
Heart rate	84–90 bpm (third-trimester baseline); simulator uses 1.40–1.50 Hz	Literature range cited within Bossung et al. 2023 (PMC10385491)	Sourced — real literature range
Breathing	~16 breaths/min (0.27 Hz), held fixed across scenarios	Al Zhranei et al. 2024 (PMID 38863888) — no significant difference in slow vital capacity, sitting vs. standing, $p=0.24$	Sourced justification for not scaling by posture; rate itself not directly measured
Fetal movement	Strong 42.9% / Moderate 39.9% / Quiet 17.3% (sitting quietly, reference category)	Bradford et al. 2019, PLOS ONE (N=266)	Sourced — controlled comparison, sitting as reference posture
Muscle noise	No quantitative magnitude; direction only (standing > sitting > lying)	Gilleard et al. 2002 (trunk ROM); transversus abdominis and pelvic-floor baseline-activity studies (non-pregnant)	Direction supported; magnitude (<i>our choice</i>)
Electrode shift	~0.08 mV mean / ~0.26 mV max bipolar shift, gradual over ~40–50s (raw, unfiltered)	Measured directly from ice002_p_1of3 (Icelandic EHG DB), the annotated “moved onto left side” event	Empirical, single real event (n=1); largely attenuated by the 0.2–3 Hz bandpass filter

Reading this table

Two mechanisms — heart rate and fetal movement — have real quantitative ranges from controlled published studies. Breathing has a real citation, but it justifies keeping the rate fixed rather than providing a seated-specific number. Muscle noise has directional support with no sourced magnitude. Electrode shift has an empirically measured value from the project's own dataset rather than a literature citation — a single real event, not a statistical distribution.

Full citations

Bossung V, Singer A, Ratz T, Rothenühler M, Leeners B, Kimmich N. "Changes in Heart Rate, Heart Rate Variability, Breathing Rate, and Skin Temperature throughout Pregnancy and the Impact of Emotions—A Longitudinal Evaluation Using a Sensor Bracelet." PMID: PMC10385491.

Al Zhranei R, Alsulami S, Alfaydi W, Alzahrani R, Alsulami M, Al Nufaiei ZF. "The impact of body position on vital capacity among pregnant women in the second trimester." Front Med (Lausanne). 2024 May 27;11:1351681. PMID: 38863888.

Bradford BF, Cronin RS, McKinlay CJD, Thompson JMD, Mitchell EA, Stone PR, McCowan LME. "A diurnal fetal movement pattern: Findings from a cross-sectional study of maternally perceived fetal movements in the third trimester of pregnancy." PLoS ONE. 2019;14(6):e0217583.

Gilleard W, Crosbie J, Smith R. "Effect of pregnancy on trunk range of motion when sitting and standing." Acta Obstet Gynecol Scand. 2002 Nov;81(11):1011-20. PMID: 12421168.

Icelandic 16-electrode Electrohysterogram Database, PhysioNet, record ice002_p_1of3 — header comment: “The participant moved onto her left side at 00:25:50 due to nausea and faintness.” Signal measured directly from the record's .dat file.

Also cited during sourcing, for context or ruled out as not directly applicable: Schneider & Deckardt 1991 (J Perinat Med, standing heart-rate acceleration, informs the standing-forecast perturbation separately from this seated table); Zhao et al. 2015 (BMJ Open, pediatric supine-to-standing HR — not used, wrong population/posture comparison).